Homework Tutor

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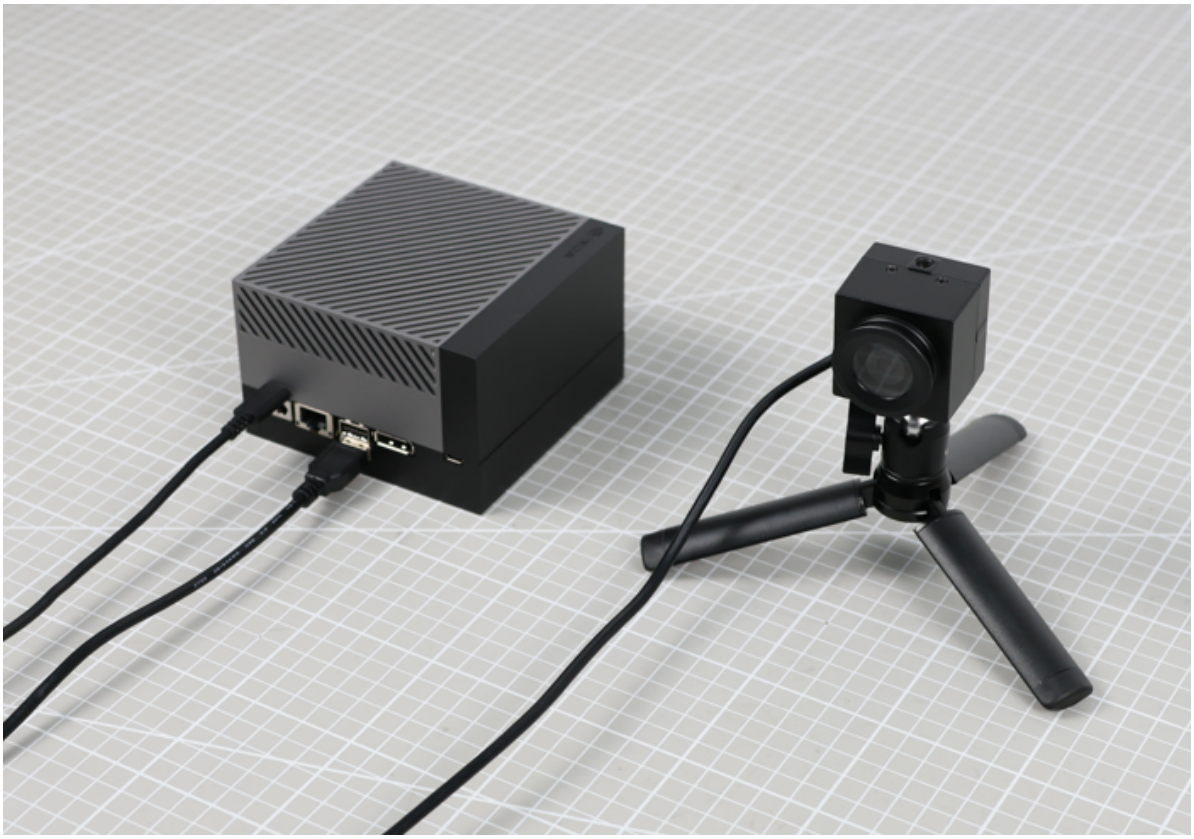
1. Function Introduction

This example uses a camera to photograph elementary or middle school homework questions. AI then generates a detailed analysis of the key points and reasoning process, finally summarizing and outputting the correct answer.

2. Hardware Connection and Setup

This example requires connecting not only a power adapter, but also a USB camera, a DP monitor, and a mouse and keyboard for displaying the interface and operation.



Open the system terminal and enter the following command. You will need to check the camera device ID.

```
ls /dev/video*
```

3. Code Analysis

Source Code Path:

```
/home/jetson/showing/homework_tutor/homework_tutor.py
```

If you need to replace the image, please place the image in the same directory as the program file and modify the IMAGE_NAME variable.

```
IMAGE_NAME = "image.jpg"
```

If you need to modify the prompt, you can modify the PROMPT variable.

```
PROMPT = "Please answer the question correctly, first analyze the key information of the question, then give an accurate and detailed reasoning process, and finally summarize."
```

If you need a model file, you can modify the MODEL_NAME variable. The selected model must support image understanding capabilities.

```
MODEL_NAME = "qwen3-vl"
```

If the problem is complex and the calculation times out, you can modify the MODEL_TIMEOUT variable. The unit is seconds, and the default is 120 seconds.

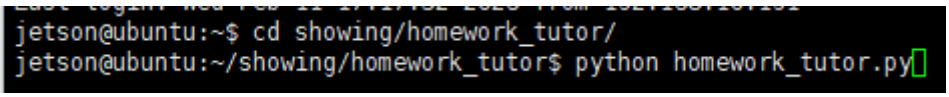
```
MODEL_TIMEOUT = 120
```

4. Running the Code

Since the program calls ollama, first ensure that the ollama service is running normally. The factory image has ollama service started by default on boot.

Open the system terminal and enter the following command to start the program.

```
cd showing/homework_tutor/
python homework_tutor.py
```



5. Experimental Results

After the program starts, it will automatically open the camera. Point the camera at the complete problem, then press the space bar. The program will capture an image and name it image.jpg, then pass it to the large model to start reasoning. The output results include key analysis of the problem, detailed reasoning process, and the final correct answer.